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SHIFT DEVICE

Abstract

In a shift device, a P position, an R position, an N position, and a D position are set on a housing, and when an electrostatic sheet detects a touch operation on the P position, the R position, the N position, or the D position and a photosensor detects a pressing operation on the housing, a shift range of a transmission is changed. Therefore, the number of components can be reduced.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to a shift device in which an operation body is operable.

BACKGROUND ART

[0002] In a shift operation switch device described in Japanese Patent Application Laid-Open No. 2012-216113, a plurality of push buttons are provided in a switch unit, and the push buttons are pushed to change a range of a vehicle.

[0003] Here, in such a shift operation switch device, the push button is provided for each range of the vehicle, and a spring corresponding to the push operation is provided for each push button.

SUMMARY OF INVENTION

Technical Problem

[0004] The present disclosure has been made in view of the above circumstance, and an object of the disclosure is to obtain a shift device capable of reducing the number of components.

Solution to Problem

[0005] According to a first aspect of the present disclosure, a shift device includes: an operation body on which a plurality of shift positions are set; and a detection portion that detects a touch operation on the shift position and a pressing operation on the operation body to change a shift range of a transmission of a vehicle.

[0006] According to a second aspect of the disclosure, in the shift device according to the first aspect of the disclosure, the plurality of shift positions are set on the same surface of the operation body.

[0007] According to a third aspect of the disclosure, in the shift device according to the first or second aspect of the disclosure, the detection portion detects the touch operation on the shift position and then detects the pressing operation on the operation body to change the shift range.

[0008] According to a fourth aspect of the disclosure, the shift device according to any one of the first to third aspects of the disclosure further includes a restriction mechanism that restricts the pressing operation on the operation body at a predetermined opportunity.

Advantageous Effects of Invention

[0009] In the shift device according to the first aspect of the disclosure, the plurality of shift positions are set on the operation body, and the detection portion detects the touch operation on the shift position and the pressing operation on the operation body to change the shift range of the transmission of the vehicle. Therefore, the number of operation bodies can be reduced, and since it is sufficient if a component corresponding to the pressing operation on the operation body is provided, the number of components can be reduced.

[0010] In the shift device according to the second aspect of the disclosure, the plurality of shift positions are set on the same surface of the operation body. Therefore, a touch operation can be easily performed on the plurality of shift positions.

[0011] In the shift device according to the third aspect of the disclosure, the detection portion detects the touch operation on the shift position and then detects the pressing operation on the operation body to change the shift range. Therefore, the operation on the operation body can be easily performed.

[0012] In the shift device according to the fourth aspect of the disclosure, the restriction mechanism restricts the pressing operation on the operation body at a predetermined opportunity. Therefore, the change of the shift range can be restricted.

Description

BRIEF DESCRIPTION OF DRAWINGS
[0013] FIG. **1** is a front view showing a shift device according to an embodiment of the present disclosure when viewed from the front.
[0014] FIG. **2** is a cross-sectional view showing the shift device according to the embodiment of the disclosure when viewed from below.
[0015] FIG. **3** is a cross-sectional view of a shift device according to a modification of the embodiment of the disclosure when viewed from below.
DESCRIPTION OF EMBODIMENTS
[0016] FIG. **1** is a front view of a shift device **10** according to an embodiment of the present disclosure when viewed from the front, and FIG. **2** is a cross-sectional view of the shift device **10** when viewed from below. In the drawings, a front side of the shift device **10** is indicated by an arrow SF, a right side of the shift device **10** is indicated by an arrow RH, and an upper side of the shift device **10** is indicated by an arrow UP.
[0017] As illustrated in FIGS. **1** and **2**, the shift device **10** according to the present embodiment is installed at the center of an instrument panel **12** of a vehicle (automobile) in a vehicle width direction, and is disposed inside a steering wheel (not illustrated) of the vehicle in the vehicle width direction. The front side, the right side, and the upper side of the shift device **10** are directed to a rear side, a right side, and an upper side of the vehicle, respectively.
[0018] The shift device **10** includes a case **14** having a substantially rectangular parallelepiped box shape as an installation body, and the case **14** is fixed to a vehicle front side of the instrument panel **12**. The case **14** is elongated in a left-right direction, and the inside of the case **14** is opened to a front side. In the instrument panel **12**, a rectangular opening **12**A is formed on a vehicle rear side of the case **14**, and the case **14** faces the opening **12**A. A fixed column **14**A having a substantially rectangular parallelepiped shape is integrally provided on a right wall of the case **14**, and the fixed column **14**A protrudes rightward and forward from the right wall of the case **14**.
[0019] A substantially rectangular parallelepiped box-shaped housing **16** as an operation body is provided on the front side of the case **14**, and the housing **16** is elongated in the left-right direction and made of, for example, a resin to be an insulator. A front housing **16**A having a substantially rectangular parallelepiped box shape is provided in a front side portion of the housing **16**, and a portion other than an outer peripheral portion of a back wall of the front housing **16**A is opened. A back housing **16**B having a substantially rectangular plate shape is provided in a back side portion of the housing **16**, and the back housing **16**B is connected to an outer peripheral portion of the back wall of the front housing **16**A. The back housing **16**B is disposed in the front housing **16**A, and the back housing **16**B is disposed parallel to a front wall of the front housing **16**A. On a right surface of the housing **16**, a restricted groove **16**C as a restricted portion is formed at a back side portion, and the restricted groove **16**C is opened to the right side and the back side.
[0020] A rectangular frame-shaped rubber **18** as a biasing member is provided between a side wall of the case **14** and the outer peripheral portion of the back wall of the housing **16**, and the rubber **18** connects the case **14** and the housing **16**. The rubber **18** is elastically contractible, and the housing **16** is displaceable backward against an elastic force (biasing force) of the rubber **18**.
[0021] A front wall of the housing **16** is fitted to the opening **12**A of the instrument panel **12**, and the flat front side surface of the housing **16** forms a design surface of a vehicle interior of the vehicle together with a vehicle rear side surface of the instrument panel **12**. Therefore, the front side surface of the housing **16** can be touched and pressed by a finger F of an occupant (particularly

a driver) of the vehicle.

[0022] A rectangular P position **20**P (park position), an R position **20**R (reverse position), an N position **20**N (neutral position), and a D position **20**D (drive position) as shift positions are set on the front side surface of the housing **16**, and the P position **20**P, the R position **20**R, the N position **20**N, and the D position **20**D are arranged at equal intervals in this order from the left to the right. In addition, a “P” indicator **16**P, an “R” indicator **16**R, an “N” indicator **16**N, and a “D” indicator **16**D are displayed at central portions of the P position **20**P, the R position **20**R, the N position **20**N, and the D position **20**D, respectively.

[0023] A rectangular sheet-shaped electrostatic sheet **22** as a touch detection portion that forms a detection portion is fixed to a back side of the front wall of the housing **16**, and the electrostatic sheet **22** covers a back side surface of the front wall of the housing **16**. A P electrode **24**P, an R electrode **24**R, an N electrode **24**N, and a D electrode **24**D each having a rectangular film shape as shift detection portions are provided on a front side of the electrostatic sheet **22**, and the P electrode **24**P, the R electrode **24**R, the N electrode **24**N, and the D electrode **24**D are arranged at equal intervals in this order from the left to the right. The entirety of each of the P electrode **24**P, the R electrode **24**R, the N electrode **24**N, and the D electrode **24**D faces the entirety of each of the P position **20**P, the R position **20**R, the N position **20**N, and the D position **20**D in a front-back direction (a thickness direction of the front wall of the housing **16**), and the P electrode **24**P, the R electrode **24**R, the N electrode **24**N, and the D electrode **24**D are made of, for example, metal and have conductivity.

[0024] The front wall (including the electrostatic sheet **22**) of the housing **16** can transmit light at portions of the “P” indicator **16**P, the “R” indicator **16**R, the “N” indicator **16**N, and the “D” indicator **16**D.

[0025] A circuit board **40** is provided in the housing **16**, and the circuit board **40** is fixed to a front side of the back housing **16**B. A P element **42**P, an R element **42**R, an N element **42**N, and a D element **42**D as illumination units are fixed to a front side surface of the circuit board **40**, and the P element **42**P, the R element **42**R, the N element **42**N, and the D element **42**D are light emitting diodes (LEDs), and face the portions of the “P” indicator **16**P, the “R” indicator **16**R, the “N” indicator **16**N, and the “D” indicator **16**D on the front wall of the housing **16** in the front-back direction, respectively.

[0026] A photosensor **26** as a pressing detection portion that forms a detection portion is fixed in the case **14**, and the photosensor **26** faces the back housing **16**B of the housing **16**. The photosensor **26** can irradiate the back housing **16**B with light and can receive light reflected by the back housing **16**B, and the photosensor **26** can detect a distance to the housing **16** and detect a backward displacement distance of the housing **16**.

[0027] A solenoid **28** as a restriction mechanism is fixed to a front side of the fixed column **14**A of the case **14**. The solenoid **28** includes a columnar plunger **28**A as a restricting portion, and the plunger **28**A faces the restricted groove **16**C of the housing **16** on the left side. When the solenoid **28** is actuated, the plunger **28**A extends leftward from the solenoid **28** and is inserted into the restricted groove **16**C, so that the plunger **28**A restricts backward displacement of a front side surface of the restricted groove **16**C, thereby restricting the backward displacement of the housing **16**.

[0028] The P electrode **24**P, the R electrode **24**R, the N electrode **24**N, and the D electrode **24**D of the electrostatic sheet **22**, the P element **42**P, the R element **42**R, the N element **42**N, and the D element **42**D of the circuit board **40**, the photosensor **26**, and the solenoid **28** are electrically connected to the control device **30** of the vehicle. The control device **30** is provided with a microcomputer in which a central processing unit (CPU), a read only memory (ROM), a random access memory (RAM), a non-volatile memory (storage), and the like are connected by a bus. In the control device **30**, the CPU reads a program stored in the ROM and the storage and executes the program while loading the program in the RAM, thereby implementing various functions. In addition, a transmission **32** (automatic transmission) and a brake **34** (foot brake) of the vehicle, a speedometer **36** of the vehicle, and an N switch **38** (neutral lock switch) are electrically connected to the control device **30**.

[0029] Next, actions according to the embodiment will be described.

[0030] In the shift device **10** having the above configuration, when the occupant performs a touch operation on the P position **20**P, the R position **20**R, the N position **20**N, or the D position **20**D on the front side surface of the housing **16** with the finger F, an electrostatic capacitance is generated between the finger F and the P electrode **24**P, the R electrode **24**R, the N electrode **24**N, or the D electrode **24**D of the electrostatic sheet **22**. Further, when the occupant presses the front side surface of the housing **16** with the finger F, the housing **16** is displaced backward against the biasing force of the rubber **18**. When the capacitance between the finger F and the P electrode **24**P, the R electrode **24**R, the N electrode **24**N, or the D electrode **24**D is equal to or larger than a capacitance threshold, and the backward displacement distance of the housing **16** is equal to or larger than a distance threshold (the backward displacement distance of the housing **16** detected by the photosensor **26** is equal to or larger than the distance threshold), the control device **30** determines that the P position **20**P, the R position **20**R, the N position **20**N, or the D position **20**D is selected, and a shift range of the transmission **32** is changed to a corresponding P range (park range), R range (reverse range), N range (neutral range), or D range (drive range) under the control of the control device **30**.

[0031] When the shift range of the transmission **32** is changed to the P range, the R range, the N range, or the D range, the P element **42**P, the R element **42**R, the N element **42**N, or the D element **42**D of the circuit board **40** emits light under the control of the control device **30**, such that the “P” indicator **16**P, the “R” indicator **16**R, the “N” indicator **16**N, or the “D” indicator **16**D on the front side surface of the housing **16** is illuminated.

[0032] In a case where the brake **34** is not operated (turned off) when the transmission **32** is in the P range (predetermined opportunity), the solenoid **28** is operated under the control of the control device **30**, and the plunger **28**A of the solenoid **28** is inserted into the restricted groove **16**C of the housing **16**, whereby the backward displacement of the housing **16** is restricted. Therefore, the change of the shift range of the transmission **32** is restricted.

[0033] On the other hand, in a case where the brake **34** is operated (turned on) when the transmission **32** is in the P range, the operation of the solenoid **28** is deactivated under the control of the control device **30**, and the plunger **28**A is separated from the restricted groove **16**C, so that the backward displacement of the housing **16** is permitted. Therefore, the change of the shift range of the transmission **32** is permitted.

[0034] In a case where a forward speed or backward speed of the vehicle is equal to or higher than a predetermined speed (a forward speed or backward speed of the vehicle measured by the speedometer **36** is equal to or higher than a predetermined speed) when the transmission **32** is in the D range or the R range (predetermined opportunity), the solenoid **28** is operated under the control of the control device **30** to restrict the backward displacement of the housing **16**. Therefore, the change of the shift range of the transmission **32** is restricted.

[0035] In a case where the forward speed or backward speed of the vehicle is lower than the predetermined speed (the forward speed or backward speed of the vehicle measured by the speedometer **36** is lower than the predetermined speed) when the transmission **32** is in the D range or the R range (predetermined opportunity), the operation of the solenoid **28** is deactivated under the control of the control device **30** to permit the backward displacement of the housing **16**. Therefore, the change of the shift range of the transmission **32** is permitted.

[0036] In a case where the N switch **38** is turned on when the transmission **32** is in the N range (predetermined opportunity), the solenoid **28** is operated under the control of the control device **30**, and the backward displacement of the housing **16** is restricted. Therefore, the change of the shift range of the transmission **32** is restricted.

[0037] On the other hand, in a case where the N switch **38** is turned off when the transmission **32** is in the N range, the operation of the solenoid **28** is deactivated under the control of the control device **30**, and the backward displacement of the housing **16** is permitted. Therefore, the change of the shift range of the transmission **32** is permitted.

[0038] In a case where one or more fingers F of the occupant touch two or more of the P position **20**P, the R position **20**R, the N position **20**N, and the D position **20**D, and an electrostatic capacitance between the one or more fingers F and the two or more of the P electrode **24**P, the R electrode **24**R, the N electrode **24**N, and the D electrode **24**D is equal to or larger than a capacitance threshold (predetermined opportunity), the solenoid **28** is operated under the control of the control device **30**, and the backward displacement of the housing **16** is restricted. Therefore, the change of the shift range of the transmission **32** is restricted.

[0039] On the other hand, in a case where a state in which one or more fingers F of the occupant touch two or more of the P position **20**P, the R position **20**R, the N position **20**N, and the D position **20**D is released and a state in which the electrostatic capacitance between the one or more fingers F and the two or more of the P electrode **24**P, the R electrode **24**R, the N electrode **24**N, and the D electrode **24**D is equal to or larger than the capacitance threshold is released, the operation of the solenoid **28** is deactivated under the control of the control device **30** and the backward displacement of the housing **16** is permitted. Therefore, the change of the shift range of the transmission **32** is permitted.

[0040] Here, the P position **20**P, the R position **20**R, the N position **20**N, and the D position **20**D are set on the housing **16**, and the P electrode **24**P, the R electrode **24**R, the N electrode **24**N, and the D electrode **24**D detect touch operations on the P position **20**P, the R position **20**R, the N position **20**N, and the D position **20**D, respectively, and the photosensor **26** detects a pressing operation on the housing **16**, whereby the shift range of the transmission **32** is changed. Therefore, the number of housings **16** can one, and a component (rubber **18**) corresponding to the pressing operation may be provided on the housing **16**, so that the number of components can be reduced and the cost can be reduced. In addition, since one housing **16** (operation body) is provided, the appearance of the shift device **10** can be improved.

[0041] Further, after the P electrode **24**P, the R electrode **24**R, the N electrode **24**N, and the D electrode **24**D detect touch operations on the P position **20**P, the R position **20**R, the N position **20**N, and the D position **20**D, respectively, the photosensor **26** detects the pressing operation on the housing **16**, whereby the shift range of the transmission **32** is changed. Therefore, it is possible to easily operate the housing **16** with the finger F of the occupant.

[0042] Moreover, the P position **20**P, the R position **20**R, the N position **20**N, and the D position **20**D are set on the front side surface (the same surface) of the housing **16**. Therefore, for example, the occupant can perform the touch operation on the P position **20**P, the R position **20**R, the N position **20**N, and the D position **20**D by tracing the front side surface of the housing **16** with the finger F, and can easily perform the touch operation on the P position **20**P, the R position **20**R, the N position **20**N, and the D position **20**D.

[0043] In addition, the solenoid **28** is operated to restrict the backward displacement of the housing **16**. Therefore, the shift range of the transmission **32** can be restricted from being unnecessarily changed, and the occupant can clearly recognize that the change of the shift range of the transmission **32** is restricted.

Modification

[0044] FIG. **3** is a cross-sectional view of a shift device **50** according to a modification of the embodiment when viewed from below.

[0045] As illustrated in FIG. **3**, in the shift device **50** according to the present modification, a circuit board **40** is fixed to a back side surface in a case **14**, and a photosensor **26** is fixed to a front side surface of the circuit board **40**.

[0046] Columnar light guides **44** as guide members are provided between a P element **42**P, an R

element **42**R, an N element **42**N, and a D element **42**D of the circuit board **40** and portions of a “P” indicator **16**P, an “R” indicator **16**R, an “N” indicator **16**N, and a “D” indicator **16**D on a front wall of a housing **16**, and the light guides **44** penetrate through and are fixed to a back housing **16**B of the housing **16**. The light guides **44** are arranged parallel to a front-back direction, and the light guides **44** are made transparent to guide light.
[0047] A P electrode **24**P, an R electrode **24**R, an N electrode **24**N, and a D electrode **24**D of an electrostatic sheet **22**, the P element **42**P, the R element **42**R, the N element **42**N, and the D element **42**D of the circuit board **40**, a photosensor **26**, and a solenoid **28** are electrically connected to a control device **30** via the circuit board **40**.
[0048] When a shift range of a transmission **32** is changed to a P range, an R range, an N range, or a D range, the P element **42**P, the R element **42**R, the N element **42**N, or the D element **42**D of the circuit board **40** emits light under the control of the control device **30**, such that the light guide **44** guides the light to illuminate the “P” indicator **16**P, the “R” indicator **16**R, the “N” indicator **16**N, or the “D” indicator **16**D on a front side surface of the housing **16**.
[0049] In the embodiment (including the modification), the front side surface of the housing **16** is flat. However, the front side surface of the housing **16** may be a curved surface.
[0050] In the embodiment (including the modification), the restriction mechanism is the solenoid **28**, and when the solenoid **28** is operated, the plunger **28**A of the solenoid **28** is inserted into the restricted groove **16**C of the housing **16**, so that the plunger **28**A restricts the backward displacement of the front side surface of the restricted groove **16**C, thereby restricting the backward displacement of the housing **16**. However, the restriction mechanism may include a motor and a restriction member, and the motor and the restriction member may be connected by, for example, a rack and pinion mechanism. When the motor is operated and the restriction member is inserted into the restricted groove **16**C of the housing **16**, the restriction member may restrict the backward displacement of the front side surface of the restricted groove **16**C, so that the backward displacement of the housing **16** may be restricted.
[0051] Furthermore, in the embodiment (including the modification), the photosensor **26** (pressing detection portion) detects the backward displacement distance of the housing **16** by light. However, the pressing detection portion may magnetically detect the backward displacement distance of the housing **16**, and the pressing detection portion may detect the backward displacement distance of the housing **16** with a switch.
[0052] In the embodiment (including the modification), the shift devices **10** and **50** are installed on an instrument panel. However, the shift devices **10** and **50** may be installed on other parts of the vehicle (such as a console or a column cover).
[0053] The disclosure of Japanese Patent Application No. 2022-070383 filed on Apr. 21, 2022 is incorporated herein by reference in its entirety.
[0054] All documents, patent applications, and technical standards mentioned herein are incorporated herein by reference to the same extent as if each individual document, patent application, and technical standard were specifically and individually stated.

Claims

1. A shift device comprising: an operation body on which a plurality of shift positions are set; and a detection portion, the detection portion being configured to detect a touch operation on the shift position and a pressing operation on the operation body, and to change a shift range of a transmission of a vehicle.
2. The shift device according to claim 1, wherein the plurality of shift positions are set on a same surface of the operation body.
3. The shift device according to claim 1, wherein the detection portion is configured to detect the touch operation on the shift position and then to detect the pressing operation on the operation body

to change the shift range.

4. The shift device according to claim 1, further comprising a restriction mechanism, the restriction mechanism being configured to restrict the pressing operation on the operation body at a predetermined opportunity.